

TD1 Answer Sheet

Name: Elias Bejjani

Group: IF1

```
● @eliasbejj → /workspaces/esilv-marketpulse-g01-HANETeam (elias_bejjani) $ ls
ls -la
CONTRIBUTING.md README.md TEAM.md config data evidence labs readiness requirements.txt src
total 64
drwxrwxrwx+ 9 codespace root 4096 Oct 7 14:04 .
drwxr-xrwx+ 5 codespace root 4096 Oct 7 14:04 ..
drwxrwxrwx+ 8 codespace root 4096 Oct 7 14:04 .git
-rw-rw-rw- 1 codespace root 178 Oct 7 14:04 .gitignore
-rw-rw-rw- 1 codespace root 2943 Oct 7 14:04 CONTRIBUTING.md
-rw-rw-rw- 1 codespace root 10270 Oct 7 14:04 README.md
-rw-rw-rw- 1 codespace root 1455 Oct 7 14:04 TEAM.md
drwxrwxrwx+ 2 codespace root 4096 Oct 7 14:04 config
drwxrwxrwx+ 3 codespace root 4096 Oct 7 14:04 data
drwxrwxrwx+ 2 codespace root 4096 Oct 7 14:04 evidence
drwxrwxrwx+ 2 codespace root 4096 Oct 7 14:04 labs
drwxrwxrwx+ 2 codespace root 4096 Oct 7 14:04 readiness
-rw-rw-rw- 1 codespace root 133 Oct 7 14:04 requirements.txt
drwxrwxrwx+ 2 codespace root 4096 Oct 7 14:04 src
○ @eliasbejj → /workspaces/esilv-marketpulse-g01-HANETeam (elias_bejjani) $ █
```

5.3)

```
● @eliasbejj → /workspaces/esilv-marketpulse-g01-HANETeam (elias_bejjani) $ cd data
pwd
ls

cd sample
pwd
ls
/workspaces/esilv-marketpulse-g01-HANETeam/data
sample
/workspaces/esilv-marketpulse-g01-HANETeam/data/sample
bloomberg_reference_expected.json bloomberg_reference_sample.json instruments.json prices.csv
○ @eliasbejj → /workspaces/esilv-marketpulse-g01-HANETeam/data/sample (elias_bejjani) $ █
```

Root)

```
/workspaces/esilv-marketpulse-g01-HANETeam/data/sample
bloomberg_reference_expected.json bloomberg_reference_sample.json instruments.json prices.csv
● @eliasbejj → /workspaces/esilv-marketpulse-g01-HANETeam/data/sample (elias_bejjani) $ cd ../../
pwd
/workspaces/esilv-marketpulse-g01-HANETeam
○ @eliasbejj → /workspaces/esilv-marketpulse-g01-HANETeam (elias_bejjani) $ █
```

6.1)

```
@eliasbejj →/workspaces/esilv-markets-pulse-g01-HANETeam (elias_bejjani) $ cat data/sample/instruments.json
{
  "instrument": {
    "ticker": "AAPL",
    "name": "Apple Inc.",
    "currency": "USD",
    "market": "NASDAQ"
  },
  "benchmark": {
    "ticker": "SP500",
    "name": "S&P 500",
    "currency": "USD",
    "market": "US"
  }
}
```

6.2)

```
@eliasbejj →/workspaces/esilv-markets-pulse-g01-HANETeam (elias_bejjani) $ head data/sample/prices.csv
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
@eliasbejj →/workspaces/esilv-markets-pulse-g01-HANETeam (elias_bejjani) $
```

AAPL)

```
@eliasbejj →/workspaces/esilv-markets-pulse-g01-HANETeam (elias_bejjani) $ grep AAPL data/sample/prices.csv
grep SP500 data/sample/prices.csv
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
2026-09-09,AAPL,254.00,255.70,252.00,254.20,38000000
2026-09-10,AAPL,254.30,256.60,253.10,255.10,39500000
2026-09-11,AAPL,254.20,256.20,253.00,254.70,41000000
2026-09-14,AAPL,256.60,258.30,255.40,256.80,42500000
2026-09-15,AAPL,257.20,259.50,256.00,258.00,44000000
2026-09-16,AAPL,256.70,258.70,255.50,257.20,38000000
2026-09-17,AAPL,259.20,260.90,258.00,259.40,39500000
2026-09-18,AAPL,259.30,261.60,258.10,260.10,41000000
2026-09-21,AAPL,259.10,261.10,257.90,259.60,42500000
2026-09-22,AAPL,261.10,262.80,259.90,261.30,44000000
2026-09-23,AAPL,261.20,263.50,260.00,262.00,38000000
2026-09-24,AAPL,263.00,265.00,261.80,263.50,39500000
2026-09-25,AAPL,262.60,264.30,261.40,262.80,41000000
2026-09-28,AAPL,263.30,265.60,262.10,264.10,42500000
2026-09-29,AAPL,264.50,266.50,263.30,265.00,44000000
2026-09-30,AAPL,266.00,267.70,264.80,266.20,38000000
```

S&P500)

```
2026-09-30,AAPL,266.00,267.70,264.80,266.20,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
2026-09-08,SP500,6623.00,6643.00,6611.00,6628.00,0
2026-09-09,SP500,6640.00,6657.00,6628.00,6642.00,0
2026-09-10,SP500,6642.00,6665.00,6630.00,6650.00,0
2026-09-11,SP500,6642.00,6662.00,6630.00,6647.00,0
2026-09-14,SP500,6663.00,6680.00,6651.00,6665.00,0
2026-09-15,SP500,6664.00,6687.00,6652.00,6672.00,0
2026-09-16,SP500,6663.00,6683.00,6651.00,6668.00,0
2026-09-17,SP500,6683.00,6700.00,6671.00,6685.00,0
2026-09-18,SP500,6684.00,6707.00,6672.00,6692.00,0
2026-09-21,SP500,6683.00,6703.00,6671.00,6688.00,0
2026-09-22,SP500,6703.00,6720.00,6691.00,6705.00,0
2026-09-23,SP500,6704.00,6727.00,6692.00,6712.00,0
2026-09-24,SP500,6715.00,6735.00,6703.00,6720.00,0
2026-09-25,SP500,6713.00,6730.00,6701.00,6715.00,0
2026-09-28,SP500,6720.00,6743.00,6708.00,6728.00,0
2026-09-29,SP500,6729.00,6749.00,6717.00,6734.00,0
2026-09-30,SP500,6740.00,6757.00,6728.00,6742.00,0
@eliasbejj →/workspaces/esilv-marketpulse-g01-HANETeam (elias_bejjani) $ █
```

Part 7)

```
@eliasbejj →/workspaces/esilv-marketpulse-g01-HANETeam (elias_bejjani) $ python --version
Python 3.14.2
@eliasbejj →/workspaces/esilv-marketpulse-g01-HANETeam (elias_bejjani) $ git --version
git version 2.55.0
@eliasbejj →/workspaces/esilv-marketpulse-g01-HANETeam (elias_bejjani) $ python src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
@eliasbejj →/workspaces/esilv-marketpulse-g01-HANETeam (elias_bejjani) $ █
```

Optional)

```
• @eliasbejj → /workspaces/esilv-marketpulse-g01-HANETeam (elias_bejjani) $ grep -c AAPL data/sample/prices.csv
grep -c SP500 data/sample/prices.csv
21
21
• @eliasbejj → /workspaces/esilv-marketpulse-g01-HANETeam (elias_bejjani) $ ls -lh
head -n 5 data/sample/prices.csv
tail -n 5 data/sample/prices.csv
total 48K
-rw-rw-rw- 1 codespace root 2.9K Oct 7 14:04 CONTRIBUTING.md
-rw-rw-rw- 1 codespace root 11K Oct 7 14:04 README.md
-rw-rw-rw- 1 codespace root 1.5K Oct 7 14:04 TEAM.md
drwxrwxrwx+ 2 codespace root 4.0K Oct 7 14:04 config
drwxrwxrwx+ 3 codespace root 4.0K Oct 7 14:04 data
drwxrwxrwx+ 2 codespace root 4.0K Oct 7 14:04 evidence
drwxrwxrwx+ 2 codespace root 4.0K Oct 7 14:04 labs
drwxrwxrwx+ 2 codespace root 4.0K Oct 7 14:04 readiness
-rw-rw-rw- 1 codespace root 133 Oct 7 14:04 requirements.txt
drwxrwxrwx+ 2 codespace root 4.0K Oct 7 14:04 src
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-28,SP500,6720.00,6743.00,6708.00,6728.00,0
2026-09-29,AAPL,264.50,266.50,263.30,265.00,44000000
2026-09-29,SP500,6729.00,6749.00,6717.00,6734.00,0
2026-09-30,AAPL,266.00,267.70,264.80,266.20,38000000
2026-09-30,SP500,6740.00,6757.00,6728.00,6742.00,0
```

Interpretation:

ls and -lh checks the folder itself to give us metadata like file sizes and permissions.

Meanwhile, head and tail actually read the contents of prices.csv to print out the first and last 5 rows of our dataset.